



## Homework-5 RMC

1)

9.2)  $m=2, b=6, k=4$

Initial position  $\rightarrow x(0) = 1$

Char equation:

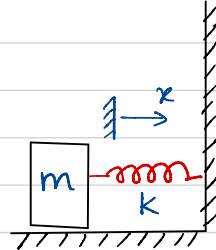
$$ms^2 + bs + k = 0$$

$$2s^2 + 6s + 4 = 0$$

$$s^2 + 3s + 2 = 0$$

$$s(s+2) + (s+2) = 0 \rightarrow (s+1)(s+2) = 0.$$

$$\therefore s = -1 \text{ or } -2.$$



Response has the form,

$$x(t) = c_1 e^{-t} + c_2 e^{-2t}$$

At  $t=0$ ,

$$x(0) = c_1 + c_2 = 1$$

$$\text{i.e. } c_1 = 1 - c_2 \text{ — ①}$$

we know that,  $x(0) = 0$

$$0 = c_1 - 2c_2$$

$$-1 + c_2 - 2c_2 = 0.$$

$$-1 - c_2 = 0$$

$$c_2 = -1 \text{ \& } c_1 = 2$$

$$x(t) = 2e^{-t} + (-1)e^{-2t}$$

9.3)  $m=1, b=2, K=1$

initial position  $x(0)=4$

char equation is,

$$ms^2 + bs + K = 0.$$

$$s^2 + 2s + 1 = 0$$

$$(s+1)(s+1) = 0$$

$$s_1 = -1 \text{ \& } s_2 = -1$$

Response has the form, (for real & equal roots)

$$x(t) = c_1 e^{s_1 t} + c_2 e^{s_2 t}$$

At  $t=0$ ,

$$c_1 = 4.$$

$$x(t) = s_1 c_1 e^{s_1 t} + c_2 [s_2 t e^{s_2 t} + e^{s_2 t}]$$

$$\text{i.e. } 0 = -c_1 + c_2 \rightarrow c_2 = c_1$$

$$\therefore c_2 = 4$$

$$x(t) = 4e^{-t} + 4te^{-t}$$

$$\Rightarrow 4e^{-t}(1+t)$$

2)

9.8)  $m=1, b=4, K=5$

$$w_{res} = 6 \text{ rad s}^{-1}$$

control law eq<sup>n</sup>:  $ms^2 + (b+Kv)s + (K+Kp) = 0$  — (1)

$$s^2 + 2zw_n s + w_n^2 \text{ — (2)}$$

Comparing we get,

$$b + Kv = 2zw_n$$

$$4 + Kv = 6 \quad (\text{as } z=1)$$

$$Kv = 2$$

$$K + Kp = w_n^2$$

$$5 + Kp = 9$$

$$Kp = 4$$

$$2w_n = w_{res} \rightarrow w_n = 3.$$

9.21)  $m=3, b=5, K=2$

Initial position  $x(0) = 3$ .

char eq<sup>n</sup> is,

$$ms^2 + bs + K = 0$$

$$3s^2 + 5s + 2 = 0$$

$$s = \frac{-5 \pm \sqrt{25-24}}{6} \Rightarrow s = -\frac{2}{3} \text{ or } s = -1$$

Response has the form,

$$x(t) = c_1 e^{-s_1 t} + c_2 e^{-s_2 t}$$

$$x(0) = c_1 + c_2 = 3 \rightarrow c_1 = 3 - c_2 \text{ — (1)}$$

$$x(t) = 0.$$

$$\frac{2}{3} c_1 + c_2$$

$$\frac{2}{3} (3 - c_2) + c_2 = 0$$

$$2 - \frac{2}{3} c_2 + c_2 = 0 \rightarrow c_2 = -6.$$

$$\therefore c_1 = 9.$$

$$x(t) = 9e^{-2/3t} + (-6)e^{-t}.$$

9.22)

$$m=2, b=3, K=8$$

$$\omega_{res} = 6 \text{ rad s}^{-1}$$

$$\text{control law eq}^n: ms^2 + (b + Kv)s + (K + K_p) = 0 \text{ --- (1)}$$

we know that,

$$s^2 + 2\zeta\omega_n + \omega_n^2 = 0 \text{ --- (2)}$$

Comparing both equations,

$$\frac{b + Kv}{m} = 2\zeta\omega_n$$

$$\frac{3 + Kv}{2} = 6 \rightarrow Kv = 9.$$

$$2\omega_n = 6 \rightarrow \omega_n = 3.$$

$$\frac{K + K_p}{m} = \omega_n^2 \rightarrow 8 + K_p = 18 \rightarrow K_p = 10.$$

3)

10.1)

$$z = \alpha t^2 + \beta.$$

By comparing we get,

$$\alpha = 2\sqrt{\theta} + 1$$

$$\beta = 3\theta^2 - \sin\theta$$

$$z' = \theta_0 + K_w e + K_\theta e$$

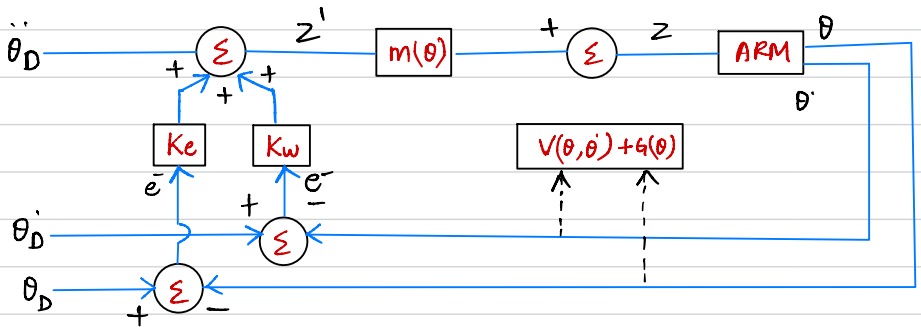
we know that,  $e = \theta_0 - \theta$ .

$$K_{CL} = K_\theta = 10.$$

$$K_w = 2\sqrt{K_{CL}} = 2\sqrt{10}$$

$$\Rightarrow K_w = 2\sqrt{10}$$

10.3)



$$K_\theta = \begin{bmatrix} K_{\theta_1} & 0 \\ 0 & K_{\theta_2} \end{bmatrix}$$

$$K_w = \begin{bmatrix} K_{w_1} & 0 \\ 0 & K_{w_2} \end{bmatrix}$$